

CHAPTER 18

Neural Control and Coordination

Neural System, Human Neural System and Neuron

1. Match List-I with List-II: (2024 Re)

List-I		List-II	
(A)	Schwann cells	(I)	Neurotransmitter
(B)	Synaptic knob	(II)	Cerebral cortex
(C)	Bipolar neurons	(III)	Myelin sheath
(D)	Multipolar neurons	(IV)	Retina

Choose the **correct** answer from the options given below:

- a. A-III, B-I, C-IV, D-II b. A-I, B-IV, C-II, D-III
c. A-IV, B-III, C-II, D-I d. A-II, B-III, C-I, D-IV

2. Match List-I with List-II: (2022 Re)

List-I		List-II	
(A)	Multipolar neuron	(I)	Somatic neural system
(B)	Bipolar neuron	(II)	Cerebral cortex
(C)	Myelinated nerve fibre	(III)	Retina of Eye
(D)	Unmyelinated nerve fibre	(IV)	Spinal nerves

Choose the **correct answer** from the options given below:

- a. A-II, B-III, C-IV, D-I b. A-III, B-I, C-IV, D-II
c. A-II, B-IV, C-III, D-I d. A-II, B-III, C-I, D-IV

3. Select the **incorrect** statement regarding synapses: (2022)

- a. Impulse transmission across a chemical synapse is always faster than that across an electrical synapse.
b. The membranes of presynaptic and postsynaptic neurons are in close proximity in an electrical synapse.
c. Electrical current can flow directly from one neuron into the other across the electrical synapse.
d. Chemical synapses use neurotransmitters

4. Nissl bodies are mainly composed of: (2018)

- a. Proteins and lipids b. DNA and RNA
c. Nucleic acids and SER d. Free ribosomes and RER

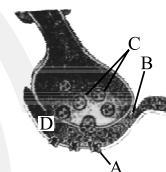
5. Myelin sheath is produced by: (2017-Delhi)

- a. Schwann Cells and Oligodendrocytes
b. Astrocytes and Schwann Cells
c. Oligodendrocytes and Osteoclasts
d. Osteoclasts and Astrocytes

6. Receptor sites for neurotransmitters are present on: (2017-Delhi)

- a. Membranes of synaptic vesicles
b. Pre-synaptic membrane
c. Tips of axons
d. Post-synaptic membrane

7. A diagram showing axon terminal and synapse is given. Identify correctly at least two of A-D (2013)



- a. C - Neurotransmitter; D - Ca^{++}
b. A - Receptor; C - Synaptic vesicles
c. B - Synaptic connection; D - K^+
d. A - Neurotransmitter; B - Synaptic cleft

Central Nervous System

8. Match List-I with List-II: (2024)

List-I		List-II	
(A)	Pons	(I)	Provides additional space for Neurons, regulates posture and balance.
(B)	Hypothalamus	(II)	Controls respiration and gastric secretions.
(C)	Medulla	(III)	Connects different regions of the brain.
(D)	Cerebellum	(IV)	Neuro secretory cells

Choose the correct answer from the options given below:

- a. A-(I), B-(III), C-(II), D-(IV)
b. A-(II), B-(I), C-(III), D-(IV)
c. A-(II), B-(III), C-(I), D-(IV)
d. A-(III), B-(IV), C-(II), D-(I)

9. Given below are two statements: (2024)

Statement-I: The cerebral hemispheres are connected by nerve tract known as corpus callosum.

Statement-II: The brain stem consists of the medulla oblongata, pons and cerebrum.

In the light of the above statements, choose the most appropriate answer from the options given below:

- a. Statement-I is correct but Statement-II is incorrect.
b. Statement-I is incorrect but Statement-II is correct.
c. Both Statement-I and Statement-II are correct.
d. Both Statement-I and Statement-II are incorrect.
10. The parts of human brain that helps in regulation of sexual behaviour, expression of excitement, pleasure, rage, fear etc. are. (2023)
a. Corpus callosum and thalamus
b. Limbic system & hypothalamus
c. Corpora quadrigemina & hippocampus
d. Brain stem & epithalamus
11. Brainstem of human brain consists of: (2023-Manipur)
a. Midbrain, Pons and Medulla Oblongata
b. Forebrain, Cerebellum and Pons
c. Thalamus, Hypothalamus and Corpora quadrigemina
d. Amygdala, Hippocampus and Corpus Callosum
12. Identify the region of human brain which has pneumotaxic centre that alters respiratory rate by reducing the duration of inspiration. (2022 Re)
a. Cerebrum b. Medulla
c. Pons d. Thalamus
13. Which part of the brain is responsible for thermoregulation? (2019)
a. Cerebrum b. Hypothalamus
c. Corpus callosum d. Medulla oblongata
14. Which of the following structures or regions is **incorrectly** paired with its function? (2018)
a. Medulla oblongata : controls respiration and cardiovascular reflexes.
b. Limbic system : consists of fibre tracts that interconnect different regions of brain; controls movement.
c. Hypothalamus : production of releasing hormones and regulation of temperature, hunger and thirst.
d. Corpus callosum : band of fibres connecting left and right cerebral hemispheres.
15. Which of the following regions of the brain is incorrectly paired with its function? (2015)
a. Corpus callosum - Communication between the left and right cerebral cortices
b. Cerebrum - Calculation and contemplation
c. Medulla oblongata - Homeostatic control
d. Cerebellum - Language comprehension
16. Injury localised to the hypothalamus would most likely disrupt: (2014)
a. Regulation of body temperature
b. Short term memory
c. Co-ordination during locomotion
d. Executive function, such as decision making

Answer Key

1. (a) 2. (a) 3. (a) 4. (d) 5. (a) 6. (d) 7. (b) 8. (d) 9. (a) 10. (b)
11. (a) 12. (c) 13. (b) 14. (b) 15. (d) 16. (a)